

Certified Temporal Frontiers for Aggregate Queries over Hidden Matchings and Privacy-Coarsened Attributes

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Abstract

Public releases often preserve one row per transaction while suppressing the key that joins rows into events and coarsening sensitive endpoint attributes. Downstream aggregate queries then depend jointly on a latent relation and the attributes attached to its endpoints. We optimize a query over feasible worlds that jointly assign a nonbipartite core–buffer matching and categorical labels subject to released supports and threshold-count factors. Independent label supports collapse to weighted matching; coupled worlds need not. We develop an exact temporal-frontier dynamic program whose state retains only live labels, matching bits, active capped-count automata, an omitted-edge budget, and a capped integer score resource. It returns attained lower and upper endpoints with witness worlds. With w live records, compiled support d , at most r active threshold factors of cap k , omission budget Γ , and integer score target b , the live-key bound is $(2d)^w [3(k+1)]^r (\Gamma+1)(b+1)$. The dependence on b is substantive: the associated threshold decision problem is weakly NP-complete even on disjoint four-cycles of pathwidth two. A downward score quantization gives certified outward query endpoints while relaxing the original score by at most $\eta|C|$; it is a bi-criteria certificate, not a query FPTAS. We use matching-level split conformal prediction as a known safety layer, with one score normalization frozen across the entire candidate-omission path. Locked exact and independent-oracle audits verify the declared finite-input semantics. A standard incidence-component layer splits only the joint record–factor graph and exactly convolves the global omission, score, and query resources; it is an engineering optimization, not a novelty claim or a universal speedup. Chicago remains a gated flagship application. A fail-closed adapter and a cross-midnight $K = 2$ production audit enforce declared handoff contracts, but public rows do not identify partner recall or run closure, and no complete-day estimate is reported.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Data provenance*; • **Theory of computation** → *Design and analysis of algorithms*.

Keywords

hidden matching, exact dynamic programming, aggregate bounds, privacy coarsening

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1 Introduction

Many public datasets preserve outcomes for individual records but remove the key identifying which records participated in the same event. Sensitive endpoint attributes may also be generalized or suppressed. Aggregate quantities such as the composition of matched pairs, referral chains, or shared-resource groups then depend on two latent objects at once: the relation and the attributes attached to its endpoints. Selecting one linkage or deleting rows with missing context creates precision by replacing released uncertainty with analyst certainty.

Propagating linkage uncertainty alone does not resolve this problem. Candidate edges can interact with latent endpoint attributes, privacy bucket counts make attribute completions mutually dependent, and the hidden matching determines which endpoint attributes enter the query. Separate link and attribute bounds can therefore combine marginally admissible choices that no single released database could have produced.

We represent the release by a set of feasible worlds. Each world jointly specifies a hidden matching and latent endpoint attributes, satisfies every released value and global coarsening constraint, and covers every core record exactly once. We compute attained lower and upper endpoints of a declared aggregate query over this set. These endpoints are conditional on the candidate supergraph and observation operator; they are not a point reconstruction, and the closed interval between them need not consist entirely of attainable values.

Learning is optional and coverage-audited. An edge model supplies a compatibility score, but node outcomes alone do not identify partner probabilities. We restrict feasible worlds by matching-level normalized regret and calibrate the radius on independent markets with observed relations. A separate supergraph budget permits selected edges outside a baseline graph. Thus learned sharpening and candidate-support sensitivity remain visible rather than being folded into one asserted graph.

Chicago ride pooling motivates the full structure. Consider a boundary-complete, internally consistent cohort with a positive shared-match flag and pooled chain size two. The public record reports realized co-presence but not the partner key. Fine geography can also be suppressed. The eventual application bounds neighborhood-context composition among those customer transactions. It does not infer rider socioeconomic status, preferences, durable ties, or all shared runs.

This version completes four bounded artifacts: joint feasible-world semantics and the independent-support matching collapse; a score-aware temporal frontier with witnesses, weak hardness, and certified outward relaxation; a provenance-bearing DNF compiler plus exact joint-incidence convolution; and locked exhaustive, all-ten UCI, and fail-closed Chicago interface audits. The component split, DNF duplication, vertex-separation identity, and conformal calibration are engineering or safety devices, not novelty

claims. Complete Chicago snapshots, verified operator semantics, partner-support evidence, and production-scale endpoints remain submission gates.

2 Related Work

Uncertain linkage and aggregate queries. Globally compatible record linkages and downstream uncertainty are established objects. Sadinle models bipartite linkage as a constrained matching [21]; Hua and Pei evaluate aggregate queries over compatible probabilistic linkages [14]; and Turkcapar and Krishnan compute extremal aggregate answers over candidate-constrained two-table linkages [27]. Attribute-annotated uncertain databases propagate value bounds through relational queries [11]. Our target difference is conditional and narrower: a within-file nonbipartite relation and endpoint attributes are simultaneously hidden, while global release counts can couple their completions. When attribute supports are independent, our problem collapses to ordinary weighted matching; that case is a baseline, not a novelty claim.

Classical cell-suppression work solves disclosure-control problems under linear table constraints by integer programming and branch-and-cut [12]. Our inverse object is different: released threshold states constrain latent micro-labels that also determine a hidden matching. Integer programming remains a general fallback, not an algorithmic novelty.

Resource-constrained matching and structured inference. An additive score floor introduces the same second resource studied in standard budgeted matching, which is NP-hard and admits PTAS/EPTAS algorithms [2, 10]. We do not assume those schemes transfer to mandatory core cover, and a generic score constraint or approximation scheme is not our contribution. Likewise, bounded-width constraint processing and bucket elimination are standard [9, 13]. Global side constraints must be compiled before width is assessed: rainbow matching remains APX-complete even when the underlying edge-coloured graph is a path [16]. Our algorithmic increment is an executable temporal state for this release-coupled world model, with active factor lifetimes, threshold caps, score-resource tracking, and endpoint witnesses.

Weak supervision and calibrated restrictions. Multiple-instance learning recovers instance-level structure from bag labels only through modeling assumptions [18]. In our application, node match outcomes do not nonparametrically identify partner-edge rankings, so we treat any edge model as a scorer. Split conformal calibration supplies finite-sample marginal coverage under exchangeability [17]. More directly, Cauchois et al. develop conformal prediction with weak supervision for structured outputs and instantiate bipartite perfect matchings, partial-matching feedback, and matching-oracle implementations [3]. Thus neither matching-level conformal prediction nor a matching oracle inside calibration is new here. Our calibration markets reveal the complete matching; we use rank calibration as a safety layer for a release-coupled nonbipartite optimizer. Candidate and release-set eligibility remain separate assumptions.

Ride pooling and contextual composition. Shareability networks and dynamic assignment solve a forward allocation problem with

observed requests [1, 22]. Chicago work predicts sharing and matching outcomes [19, 26], relates pooled demand to neighborhood disadvantage [25], and constructs proxy groups by spatiotemporal clustering [24]. Our flagship application asks an inverse measurement question: among a precisely defined set of realized two-transaction matches, which neighborhood-context compositions remain compatible with the suppressed partner key and the verified release operator? It does not infer rider socioeconomic status, preferences, durable ties, or an echo chamber.

3 Problem and Query Semantics

Released records and roles. Let V index released records and let $G = (V, E)$ be a declared candidate supergraph. Core records $C \subseteq V$ belong to the target population; boundary-buffer records $B \subseteq V$ may be partners of core records; and context-only records $U = V \setminus (C \cup B)$ enter privacy counts but cannot be selected as partners. A matching $M \subseteq E$ must satisfy

$$\deg_M(i) = 1 \quad (i \in C), \quad \deg_M(i) \leq 1 \quad (i \in B), \quad (1)$$

and contains no buffer–buffer edge. This boundary rule lets a core record use a counterpart immediately outside the target window without forcing every buffer record into the target.

Latent attributes and release constraints. Each record i has a finite support $\mathcal{A}_i$ for a latent categorical attribute a_i . Supports may be singletons for released values or larger sets for coarsened values. In the implemented reference solver, $h(i)$ assigns each record to one release cell for this modeled attribute. The verified observation operator supplies rational count bounds

$$L_{cv} \leq \sum_{i:h(i)=c} \mathbb{I}\{a_i = v\} \leq U_{cv} \quad (c, v) \in \mathcal{K}. \quad (2)$$

More specific threshold or paired-suppression logic can be compiled into the same feasible-world representation from explicit LOW/HIGH release formulas. The compiler requires an operator identifier, an audit reference, and exact row-to-factor bindings; these are provenance inputs, not proof that the external mechanism is correct. We do not infer a converse suppression rule from missingness alone.

Feasible worlds and queries. A feasible world $F = (M, a)$ satisfies Eqs. (1)–(2), $a_i \in \mathcal{A}_i$, and any declared attribute-conditioned edge rules. Let $\mathcal{F}(O, G)$ denote the finite set of such worlds. For a symmetric pair function g , define the core-transaction query

$$Q_C(F) = \frac{1}{|C|} \sum_{\{i,j\} \in M} |\{i, j\} \cap C| g(a_i, a_j). \quad (3)$$

Core–core edges contribute twice and core–buffer edges once, so the denominator remains $|C|$. Same-bin composition uses $g(a_i, a_j) = \mathbb{I}\{b(a_i) = b(a_j)\}$; signed linear or regression queries replace g by the corresponding endpoint contribution.

We report

$$\underline{Q} = \min_{F \in \mathcal{F}(O, G)} Q_C(F), \quad \overline{Q} = \max_{F \in \mathcal{F}(O, G)} Q_C(F). \quad (4)$$

PROPOSITION 1 (ENDPOINT SEMANTICS). *If $\mathcal{F}(O, G)$ is nonempty, both extrema in Eq. (4) are attained. Every feasible world’s query value lies between them. If the true world $F^\dagger$ belongs to $\mathcal{F}(O, G)$, its*

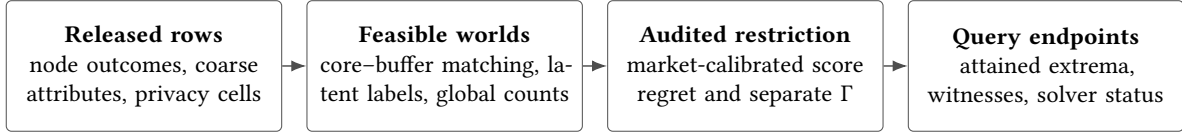


Figure 1: The method never fills in one preferred relation. The observation operator defines a joint world set; learning may restrict that set only through a separately calibrated rule; optimization then propagates all surviving ambiguity to the downstream query.

Table 1: Claim boundary induced by the four input layers.

Layer	Supplied object	Failure if misspecified
Relation support	Supergraph and core/buffer roles	True partner absent
Attribute support	Per-record finite domains	True endpoint value absent
Observation operator	Count and release factors	Inconsistent world deletion
Learned restriction	Score radius and Γ budget	Model-induced deletion

query value is covered. The proposition does not imply that every scalar between the endpoints is attainable.

PROOF. The domain is a finite set of matchings and finite attribute assignments, so a real-valued query attains its minimum and maximum. The bracketing and coverage claims follow from membership. Two feasible worlds can attain only two distinct values, so finiteness alone provides no interpolation claim. $\square$

3.1 Observation Operators as Auditable Constraints

The world set must encode what the release mechanism actually implies, not a convenient completion rule. We separate four inputs. *Record supports* translate released or coarsened values into finite domains. *Count factors* constrain assignments across every row contributing to a privacy cell, including context-only rows. *Release factors* state the verified direction of the mechanism: for example, a visible value may imply that a count cleared a threshold, while a hidden value need not imply the converse. Finally, *compatibility factors* specify which attribute pairs allow an edge. An omitted rule leaves worlds in the set; it does not silently choose a default completion.

This separation yields three different statements. The score-free endpoints are conditional on the first three layers. A learned endpoint adds the fourth. A sampling statement additionally requires a design over complete markets. None of these is a confidence interval merely because two numbers are reported. The artifact therefore stores the two endpoint witnesses and a solver-status record alongside every range.

4 Coupled Endpoint Computation

The method must distinguish a useful baseline from the coupled problem.

LEMMA 1 (INDEPENDENT-SUPPORT COLLAPSE). *Suppose (i) E does not depend on latent attributes, (ii) the attribute domain is Cartesian, $\prod_i \mathcal{A}_i$, with no cross-record count constraint, and (iii) the query is pair additive as in Eq. (3). Define*

$$\ell_{ij} = \min_{a_i \in \mathcal{A}_i, a_j \in \mathcal{A}_j} |\{i, j\} \cap C| g(a_i, a_j),$$

$$u_{ij} = \max_{a_i \in \mathcal{A}_i, a_j \in \mathcal{A}_j} |\{i, j\} \cap C| g(a_i, a_j).$$

Then the lower endpoint is a minimum-weight core-cover matching with weights $\ell_{ij}/|C|$, and the upper endpoint is the corresponding maximum-weight matching with weights $u_{ij}/|C|$.

PROOF. For any fixed matching, selected edges share no endpoint. Under a Cartesian attribute domain, each selected edge's minimizing or maximizing endpoint assignment can therefore be chosen independently and combined into one global assignment. Labels of unselected nodes can be completed arbitrarily from their nonempty supports. Optimizing the resulting edge sums over matchings gives the stated equalities. $\square$

Thus exact cover, independent missing-bin envelopes, and signed edge objectives are weighted-matching baselines. The nonseparable problem begins when release counts or attribute-conditioned feasibility prevent the edgewise choices in Lemma 1 from coexisting.

Why the joint world changes an endpoint. Consider four core records with fixed edges ab and cd . Their supports are $\{A, B\}$, $\{A, C\}$, $\{A, D\}$, and $\{A, E\}$, respectively. Let A form bin 0, B, D bin 1, and C, E bin 2. With independent supports, the same-bin query ranges from 0 to 1: choose (B, C, D, E) for the lower witness and all A for the upper witness. Add the single global release constraint $\#\{i : a_i = A\} \leq 2$. At most one edge can now have equal labels, so the attained range contracts to $[0, 1/2]$. Optimizing a matching range and an attribute range separately would retain the impossible upper value 1.

Attributes can also change the relation rather than only its query value. Let one core c have support $\{A, B\}$ and two singleton buffers $x = A$ and $y = B$. If edge cx permits only (A, A) and cy only (B, B) , a count constraint fixing $c = A$ selects x , while one fixing $c = B$ selects y . The implemented exact oracle reproduces both examples. They isolate the interaction absent from the independent-support baseline without invoking a privacy mechanism that has not been verified.

A coupled complexity boundary. Fix an integer threshold $k \geq 3$. Consider a paired row-release operator that publishes both endpoint attributes of record i exactly when both its pickup and drop-off cells contain at least k records; otherwise it publishes neither. This

is an abstract operator, not an assertion about the current Chicago release.

THEOREM 1 (PAIRED-SUPPRESSION FEASIBILITY). *For every fixed $k \geq 3$, deciding whether a feasible attribute assignment and perfect matching exist under the paired operator is NP-complete, even when the outer relation is complete bipartite, compatibility depends only on the left record's pickup attribute, and only left pickup attributes have non-singleton supports.*

PROOF SKETCH. Membership in NP follows by checking assignments, cell counts, release states, compatibility, and matching degrees. For hardness, reduce perfect three-dimensional matching on equally sized sets X, Y, Z . Each $x \in X$ chooses a latent pickup label $p_x \in Z$. For every z , add $k - 1$ visible context anchors to its pickup cell; all anchors share a separate already-large drop-off cell. Visibility forces $(k - 1) + |\{x : p_x = z\}| \geq k$. Since $|X| = |Z|$, every z is chosen exactly once. Give each x a private low-count drop-off cell so its two attributes remain paired-suppressed. Match X perfectly to Y and allow edge (x, y) under label z exactly when (x, y, z) is an input triple. Feasible worlds are therefore in bijection with perfect 3D matchings. The full construction, including the $|Z| \geq 2$ padding and size bound, is in Appendix C. $\square$

The reduction relies on paired row suppression and arbitrary tabular attribute-conditioned compatibility. The construction does not apply to endpoint-wise independent suppression and does not establish hardness for a metric-only edge rule. These boundaries must be checked, not blurred, in any application.

Compiling release formulas. For a scalar release factor f , let $N_f(a) = \sum_i \chi_{if}(a_i)$ with $\chi_{if} \in \{0, 1\}$. Its temporal automaton stores a capped count q_f and a requirement $\rho_f \in \{\emptyset, \text{LOW}, \text{HIGH}\}$. For a pure threshold k , $q_f = \min\{k, N_f^{\text{processed}}\}$; closing the factor accepts LOW iff $q_f < k$ and HIGH iff $q_f = k$. Opposing nonempty requirements conflict. General lower/upper bounds use the smallest cap that retains every decision boundary, including a distinct overflow state for a binding upper bound.

For row i , substantive label a , and terminal factor predicates T_f , the implemented interface accepts an explicit DNF

$$\Phi_i(a) = \bigvee_{j=1}^{c_i(a)} \bigwedge_{(f,r) \in C_{iaj}} [T_f(N_f) = r], \quad r \in \{\text{LOW}, \text{HIGH}\}. \quad (5)$$

It replaces a by one compiled copy (a, j) per clause, copies all edge score, query, and compatibility maps, and emits the clause literals as factor requirements. The sole empty clause is canonical TRUE. Under the abstract paired operator, visible is one HIGH-HIGH clause and hidden is the two-clause LOW-or-LOW formula.

LEMMA 2 (EXPLICIT-DNF RELEASE COMPILER). *The projection of feasible compiled worlds is exactly the release-augmented world set satisfying Eq. (5). A projected world has one preimage for each combination of its satisfied clauses. Compilation preserves the node/edge topology, the record-bag width for the same forget order, score-floor feasibility, and every label- and edge-additive query; it changes label support from $|\mathcal{A}_i|$ to $\sum_{a \in \mathcal{A}_i} c_i(a)$ and may extend factor lifetimes and thereby increase active-factor width. Counts capped at their last decision boundary remain exact.*

PROOF. For completeness, choose one satisfied clause for every row and attach its index to the substantive label. For soundness, read the clause embedded in each compiled label; accepting factor automata satisfy every emitted literal. The lifted maps copy feasibility, score, and query values across clause witnesses, so projection preserves endpoints. Compilation changes no record, edge, or action identifier. Counts on the same side of their final decision boundary have identical future transitions, so saturation is lossless. $\square$

The compiler is polynomial in the *explicit* DNF and lifted output, not a succinct arbitrary Boolean formula; exponential expansion and LOW/HIGH middle regions are exposed interface limits.

Temporal frontier. A supplied forget order is compiled into node-introduction, edge-processing, and node-forget actions. Every edge is processed while both endpoints are live; a record is forgotten only after all incident edges; and factor f is implicitly opened before its first possible contributor or requirer and closed immediately after its last. Let w be the maximum live-record count, d the largest compiled label support (including release-witness copies), B_t the live record bag, and $\mathcal{A}_t$ the active factor set at action t . Factor f retains counts $0, \dots, K_f$.

For allowed label-pair scores $s_e(u, v)$, set

$$a = \min \left\{ 0, \min_{e, (u,v) \in \mathcal{P}_e} s_e(u, v) \right\}.$$

Every world has total selected core incidence $N = |C|$, so replacing s_e by $s_e - a \geq 0$ adds the same $-aN$ to every score. Clear rational denominators and cap the accumulated integer score at the transformed floor b .

THEOREM 2 (EXACT SCORE-AWARE TEMPORAL-FRONTIER ENDPOINTS). *For a valid schedule, omission budget Γ , rational additive query, and score floor, the temporal-frontier dynamic program returns infeasible exactly when the declared world set is empty; otherwise it returns both attained query endpoints and witness worlds, provided the declared frontier-record limit is not exceeded. Exceeding that operational limit raises an explicit frontier-limit exception and licenses no feasibility conclusion. Before dominance pruning, the number of live keys at action t are at most*

$$\left(\prod_{i \in B_t} m_i d'_i \right) (\Gamma + 1)(b + 1) \prod_{f \in \mathcal{A}_t} 3(K_f + 1), \quad (6)$$

where $d'_i = \sum_{a \in \mathcal{A}_i} c_i(a)$, $m_i = 2$ for matchable records and $m_i = 1$ for context-only records. If $d'_i \leq d$, $|B_t| \leq w$, all active pure thresholds have cap k , and at most r are active, this is at most $(2d)^w [3(k + 1)]^r (\Gamma + 1)(b + 1)$. Without a score floor the factor $(b + 1)$ disappears. The work per transition is polynomial in the explicit state and input size.

PROOF SKETCH. At action t , a key stores live labels, live matching bits, the (q_f, ρ_f) automaton for each $f \in \mathcal{A}_t$, used omitted edges, and the score capped at b . For each key the program keeps separate exact minimum and maximum accumulated query values. These fields contain every processed quantity on which a future transition can depend. The running-intersection conditions ensure that forgotten labels and finalized factors can never be consulted again. Induction over actions therefore shows that the retained

values are exactly the extrema among partial worlds inducing the key. Terminal paths have every core matched, every factor finalized, and score state b ; they project onto precisely the feasible full worlds. Stored choices recover the two witnesses. Counting each live label, matching bit, factor automaton, and global resource gives Eq. (6). $\square$

For a forget order $\pi = (v_1, \dots, v_n)$, define the forward boundary $\partial_\pi^+(i) = \{v_j : j > i, \exists h \leq i : \{v_h, v_j\} \in E\}$. The compiler's largest record bag is exactly $1 + \max_i |\partial_\pi^+(i)|$, the reverse-order vertex-separation number plus one; minimizing over orders yields pathwidth [15]. Hence, if records are ordered by nondecreasing time and every candidate edge spans at most Δ , then $w \leq \max_t \{i : t \leq \tau_i \leq t + \Delta\}$. Active-factor width is likewise the maximum overlap of the compiler-audited factor lifecycle intervals. These are data-measurable certificates for a supplied order, not a claim that the order minimizes pathwidth.

An exact algorithm polynomial in binary input length at fixed structural width would imply $P = NP$; our dynamic program pays a pseudo-polynomial score coordinate.

THEOREM 3 (WEAK HARDNESS AT PATHWIDTH TWO). *For non-negative integer score and query contributions encoded in binary, deciding whether a perfect matching has score at least B and additive query at most B is weakly NP-complete even with singleton labels, no release factors, $\Gamma = 0$, and a candidate graph that is a disjoint union of four-cycles.*

PROOF SKETCH. Reduce SUBSET SUM. For item a_i , a four-cycle has two perfect matchings, “take” and “skip”. Give one take edge score a_i per core incidence and query $2a_i$; all other contributions are zero. With both thresholds $2B$, a global matching is feasible exactly when the taken items sum to B . Each component, and their concatenation, has pathwidth two. Membership in NP is immediate. This is the model's pseudo-polynomial boundary, consistent with known budgeted-matching hardness and approximation results [2, 10]. $\square$

Certified score relaxation. When b is too large, choose rational $\eta > 0$, set $r_e = \lfloor (s_e - a)/\eta \rfloor$, and impose

$$\widehat{b} = \max\{0, \lceil (B - Na)/\eta \rceil - N\}.$$

PROPOSITION 2 (OUTWARD SCORE CERTIFICATE). *If $\mathcal{F}_B$ is the exact score-floor set and $\widehat{\mathcal{F}}_\eta$ the rounded set, then*

$$\mathcal{F}_B \subseteq \widehat{\mathcal{F}}_\eta \subseteq \mathcal{F}_{B-\eta N}. \quad (7)$$

Consequently, optimizing over $\widehat{\mathcal{F}}_\eta$ gives outward query endpoints. Each returned witness misses the original score floor by at most ηN ; if an endpoint witness satisfies the original floor, that endpoint is exact.

PROOF. Write $c_e = |e \cap C|$ for selected-edge core incidence and suppress the selected label-pair argument of s_e and r_e . Fixed core incidence gives $\eta \sum c_e r_e \leq \sum c_e (s_e - a) < \eta (\sum c_e r_e + N)$, which proves both inclusions after integer rounding. Set inclusion gives the endpoint order. If a relaxed extremal witness is also exact-feasible, the reverse endpoint inequality follows from that witness. $\square$

Equation (7) is a bicriteria score certificate, not a query FPTAS: a tiny score slack may cause an arbitrary query jump.

Reference formulation. The implemented coupled reference program uses binary edge variables z_e , assignment variables x_{iv} , and joint edge-label variables y_{euv} only for ordered pairs (u, v) in an edge's declared compatibility set $\mathcal{P}_e$. It enforces Eq. (1), $\sum_{v \in \mathcal{A}_i} x_{iv} = 1$, the count bounds in Eq. (2), and

$$\begin{aligned} \sum_{(u,v) \in \mathcal{P}_e} y_{euv} &= z_e, \\ y_{euv} &\leq x_{iu}, \quad y_{euv} \leq x_{jv}, \\ y_{euv} &\geq z_e + x_{iu} + x_{jv} - 2. \end{aligned}$$

The query is linear in y . Context-only records have assignment variables and enter counts but have no incident selectable edge. An exhaustive enumerator is the exact oracle on small instances; the sparse mixed-integer program is a numerical reference backend. It also enforces optional score floors and supergraph-edge budgets. Exact optima, numerical optima, exact or structural infeasibility, numerical infeasibility, and unresolved runs are reported as distinct states; a timeout is unresolved rather than a separate certificate.

The temporal implementation accepts compiled tuple labels, label-dependent factor contributions and requirements, and label-pair-specific scores and queries. It uses exact rational arithmetic and independently replays every endpoint witness. The high-level compiler implements Lemma 2. A committed Chicago wrapper additionally validates exact node, role, and ordered-support equality, rejects undeclared or preloaded Chicago factors, freezes its hand-off, and replays a content-addressed contract before compilation. These checks certify the interface under declared inputs; they do not validate the City's implementation, a live extraction, latent-label coverage, or candidate-edge coverage. General MILP remains a numerical fallback when an exact frontier exceeds its declared limit.

PROPOSITION 3 (EXACT INCIDENCE-COMPONENT CONVOLUTION). *Form an incidence graph whose vertices are records and declared count/release factors. Candidate edges join their endpoint records, and a record joins a factor whenever any supported label can contribute to or require that factor. For its connected components, convolving the local nondominated omission, globally shifted score, and additive-query frontiers returns the same exact feasibility status and attained endpoints as the monolithic temporal solver, unless a declared local or convolution frontier limit is exceeded.*

Every structural constraint is local by construction, so restriction and union give a bijection between global worlds and tuples of component worlds. The three remaining resources are additive, making separate min/max dominance safe; unioned endpoint witnesses are replayed against the unsplit problem. Candidate-graph components alone are unsafe because a shared factor can couple otherwise disconnected records. This is standard exact decomposition followed by pseudo-polynomial knapsack-style convolution, not a new identification or complexity result.

5 Learning-Augmented Restrictions

5.1 What Node Outcomes Do Not Identify

Let $\partial M \in \{0, 1\}^V$ be the incidence vector of a matching. A released match indicator can reveal ∂M without revealing M .

Table 2: Algorithmic boundary of the declared world model.

Regime	Endpoint computation	Status
Fixed attributes	Weighted core-cover matching	Polynomial baseline
Independent supports	Two edge-envelope matchings	Lemma 1
Fixed labels plus score floor	Mandatory core cover with a budget	Related standard problem: NP-hard; PTAS/EPTAS known
Paired threshold coupling	Joint labels and matching	Theorem 1
Temporal schedule ($w, \mathcal{A}_t$)	Exact frontier DP	Theorem 2
Joint incidence components	Local frontiers plus exact convolution	Proposition 3
General connected instance	Temporal DP or numerical MILP	Width/limit/gap explicit

Table 3: Machine-readable solver states and their licensed interpretation.

State	Source	Licensed statement
EXACT_OPTIMAL	Exhaustive oracle or frontier DP	Attained endpoint certified
EXACT_INFEASIBLE	Frontier DP	Declared world set empty
OUTER_OPTIMAL	Rounded frontier DP	Outward endpoint; report score slack
NUMERICALLY_OPTIMAL	HiGHS incumbent	Numerical endpoint; retain gap
PROVEN_INFEASIBLE	Structural check or enumeration	Declared world set empty
NUMERICALLY_INFEASIBLE	HiGHS status	Numerical evidence only
UNRESOLVED	State/time limit or solver failure	No feasibility conclusion

PROPOSITION 4 (PARTNER NONIDENTIFICATION). *If a candidate graph contains distinct matchings M_1 and M_2 with $\partial M_1 = \partial M_2$, then node match outcomes and node covariates alone do not identify which of those relations occurred. In particular, an all-positive incidence vector does not distinguish any two perfect matchings.*

PROOF. Take two latent worlds with the same released nodes, covariates, and incidence vector, but relation M_1 in one and M_2 in the other. Their observations coincide while their partner edges differ. No function of the released data can distinguish them without another model or source of relation truth. $\square$

This result does not prohibit structured weak supervision; it fixes its claim boundary. A model may score complete matchings and enforce mutual exclusion, but a node-level objective alone cannot validate partner ranks in the regime of Proposition 4. In our controlled benchmark, edge scorers are therefore trained on separate synthetic source markets whose relations are observed. Chicago node outcomes are not used as hidden edge labels.

5.2 Calibrating a Matching Set

Let s_e be a matching-only scorer fixed and deterministically rationalized before calibration. With optional buffer nodes, use core incidence,

$$S_C(M) = \sum_{e \in M} |e \cap C| s_e. \quad (8)$$

Because $\sum_{e \in M} |e \cap C| = |C|$, replacing s_e by $as_e + b$ for $a > 0$ changes every feasible score by the same additive constant after scaling. For market d , let $h_d(M)$ count selected edges in $E_d^{\text{super}} \setminus E_d^{\text{base}}$ and define

$$\mathcal{F}_{d,\Gamma}^0 = \{F \in \mathcal{F}(O_d, G_d^{\text{super}}) : h_d(M(F)) \leq \Gamma\}.$$

Before calibration, fix a reference budget $\bar{\Gamma}_d$ by the same rule in every market and compute once

$$S_{d,\star}^- = \min_{F \in \mathcal{F}_{d,\bar{\Gamma}_d}^0} S_C(M(F)), \quad S_{d,\star}^+ = \max_{F \in \mathcal{F}_{d,\bar{\Gamma}_d}^0} S_C(M(F)).$$

These extrema are frozen along the entire Γ path. Define

$$R_{d,\star}(F) = \frac{S_{d,\star}^+ - S_C(M(F))}{S_{d,\star}^+ - S_{d,\star}^-}, \quad (9)$$

with value zero for a constant reference score. Recomputing the normalizer at each Γ can destroy nesting: a newly admitted high-score world can raise the floor enough to delete an old world.

For any $\tau \in [0, 1]$ and $\Gamma \leq \bar{\Gamma}_d$, define

$$\mathcal{F}_{d,\tau,\Gamma} = \left\{ F \in \mathcal{F}_{d,\Gamma}^0 : S_C(M(F)) \geq S_{d,\star}^+ - \tau(S_{d,\star}^+ - S_{d,\star}^-) \right\}. \quad (10)$$

Use m calibration markets, disjoint from scorer training, whose true matchings are observed. For score miscoverage α_S , let $k_S = \lceil (m+1)(1-\alpha_S) \rceil$ and let $\hat{\tau}_{\alpha_S}$ be the k_S th smallest calibration regret, using the support bound one if $k_S = m+1$. Retain the set in Eq. (10) at $\tau = \hat{\tau}_{\alpha_S}$. Exact additive scores parse each component through its declared decimal spelling before rational termwise summation; the same score map is used for calibration and the endpoint solver. A label-dependent scorer would instead require calibration truth for the full world, not only its matching.

THEOREM 4 (MARKET-LEVEL SCORE RETENTION AND QUERY COVERAGE). *Suppose the scorer, observation operator, graph rule, reference budgets, and normalization are fixed before calibration. Suppose the augmented markets, including their true worlds, are exchangeable, every reference set is nonempty, and $F_d^\dagger \in \mathcal{F}_{d,\bar{\Gamma}_d}^0$ almost surely. Then*

$$\Pr\{R_{m+1,\star}(F_{m+1}^\dagger) \leq \hat{\tau}_{\alpha_S}\} \geq 1 - \alpha_S.$$

For a reported $\Gamma \leq \bar{\Gamma}_{m+1}$, if $\Pr\{F_{m+1}^\dagger \notin \mathcal{F}_{m+1,\Gamma}^0\} \leq \alpha_G$, then

$$\Pr\{F_{m+1}^\dagger \in \mathcal{F}_{m+1,\hat{\tau}_{\alpha_S},\Gamma}\} \geq 1 - \alpha_S - \alpha_G.$$

On this event, every query range optimized over that same set covers its true value.

PROOF. Exchangeability gives the usual split-conformal rank bound. A union bound with the final- Γ eligibility failure gives the second display; independence is unnecessary. Conditional on true-world retention, Proposition 1 covers all queries sharing that world set, so no union bound over queries is needed. $\square$

A supplied supergraph or synthetic calibration alone establishes neither rate for Chicago. With the frozen reference range, the two sensitivities do form a nested audit rather than one opaque knob.

Table 4: Locked temporal-frontier checks; timings are not headline evidence.

Family	Controlled change	Certified outcome
Local factors	Grouped vs. interleaved	active $r : 1 \rightarrow 3$; same endpoints
Degree-one graph	Adjacent vs. nested order	schedule width 1 $\rightarrow$ 16
Binary C_4 items	pathwidth 2, $b = 2048$	peak frontier 4098
Γ path	$0 \rightarrow 8$ omitted edges	upper endpoint 0 $\rightarrow$ 8
Score relaxation	$\eta = 1/5, N = 4$	$[0, 0] \subset [-5, 0]$; deficit $2/5 \leq 4/5$

PROPOSITION 5 (NESTED SENSITIVITY PATHS). *If $\tau \leq \tau'$ and $\Gamma \leq \Gamma'$, then $\mathcal{F}_{d,\tau,\Gamma} \subseteq \mathcal{F}_{d,\tau',\Gamma'}$. Consequently, a query's lower endpoint cannot increase and its upper endpoint cannot decrease when either budget is relaxed.*

PROOF. Both structural budgets and one fixed score halfspace are nested. Optimization over the resulting superset gives the endpoint monotonicity. $\square$

Reporting the (τ, Γ) path exposes whether apparent precision comes from score or candidate exclusion; it does not turn either into graph-support evidence.

6 Controlled Evaluation

6.1 Structural and Solver Audit

The locked temporal benchmark contains 34 hashed cases. An implementation-independent rational enumerator agrees in 32/32 applicable cases, analytic answers agree in 24/24, and resolved HiGHS comparators agree numerically in 16/16. Every endpoint witness is replayed from raw inputs. Table 4 isolates the structural parameters rather than presenting wall-clock speedups. A separate legacy audit of the general MILP backend still agrees with exhaustive statuses on 250/250 random instances and endpoints on all 185 feasible instances; that agreement does not make floating-point MILP exact.

With the pinned cache, 150 tests pass: 87 bounds and frontier, 14 Chicago adapter, 23 production audit, four runtime profile, and 22 external benchmark tests. Eight executable legacy counterexamples also pass. The component layer agrees exactly in 160/160 same-kernel configurations and 192/192 independent exhaustive-oracle configurations, with 682 decomposed endpoint-witness replays. A shared-factor counterexample shows that splitting candidate-graph components can report upper endpoint 1 when the correct joint-incidence upper endpoint is 11. These checks support software behavior for declared inputs, not external operator truth or candidate coverage.

A separate 23-case capacity profile resolves 22 cases under three seconds and retains one explicit timeout. The component profile reports frontier record counts rather than memory or universal

speedups: a legal component-concatenated monolithic order remains exact, while only the supplied interleaved order exceeds its 10,000-record limit at 13 components.

6.2 External Relation-Truth Boundary

The all-ten-block UCI audit reconciles 5,749,132 unique candidate pairs and 20,931 adjudicated positives over 99,788 observed records [23]. The positive relation is not a matching: 8,675 records have positive degree above one, and its 12,925 entity components have size up to nine. The ten files are edge partitions rather than markets or folds; 82,846 records occur in every block.

An explicitly truth-conditioned reduction retains 10,297 global two-record positive components with observed true postal comparison. Their induced graph has 249,048 candidate edges and one component containing 93.94% of retained dyads. The adjudicated postal-agreement value is $9922/10297 = 0.963582$. Verified integer-weight Blossom optimization attains the exact upper endpoint $9924/10297 = 0.963776$, but the lower endpoint exceeded the predeclared 120-second limit and remains UNRESOLVED. We therefore report neither a complete interval, width, nor containment claim. The construction validates neither UCI blocking recall, independent calibration markets, nor Chicago transfer. FEBRL4 remains a small matching-method-fit check [20].

6.3 Matching-Set Calibration under Deliberate Query Leakage

Design. We use a deterministic, disjoint source/calibration/test split. Forty source markets directly supervise logistic edge scorers with observed true edges; no node-label learner is evaluated here. Forty-nine target-style markets calibrate Eq. (9), and 120 held-out markets are used for final evaluation. Each market has six hidden pairs, fully observed binary endpoint attributes, and a complete candidate graph. There is no privacy-count coupling in this experiment, so it isolates the matching-set calibration layer.

Target-free features measure route, time, and duration similarity. A query-leaking diagnostic additionally sees the same-SES edge contribution that defines the downstream query. The source homophily generating probability is 95%, while calibration and test use 55%, deliberately shifting that circular feature. The nominal matching-set coverage is 90%. Radius 0.05 is an illustrative stress point, not a selected or calibrated baseline. The committed benchmark enumerates all 10,395 perfect matchings per market using termwise decimal-rational scores and exact floor membership. A second solver-free program shares the locked generator, scorers, and rationalization utilities but separately reaggregates the 44-point frontier: discrete counts agree exactly and serialized headlines within 3.33×10^{-16} . It scores each edge once; the all-core incidence convention in Eq. (8) doubles every matching score and leaves normalized-regret sets unchanged.

Result. The query-leaking radius-0.05 stress point looks strongest if evaluated only by width: it removes 95.8% of the score-free ambiguity. It retains 18/120 true matchings and covers the downstream query in 29/120 markets. Calibration detects the shifted shortcut. Its radius expands to 0.504, retaining 117/120 true matchings and covering the scalar query in 120/120 markets, while width reduction falls to 7.6%. The target-free score needs a smaller radius but also

Table 5: Fixed held-out synthetic markets: coverage cannot be read from width alone.

Scorer	Calibrated restriction			Arbitrary radius 0.05			Point MAE
	Match cov.	Query cov.	Width red.	Match cov.	Query cov.	Width red.	
Target-free	95.8%	100%	3.8%	30.0%	83.3%	70.6%	0.122
Query-leaking diagnostic	97.5%	100%	7.6%	15.0%	24.2%	95.8%	0.308

gains little calibrated precision. This single seed and market size do not establish robustness. The result is narrower: calibration makes the coverage price of sharpening measurable rather than rewarding circular precision.

Software audit. The benchmark validates matching-level calibration only. It does not validate node weak supervision, latent-attribute inference, privacy-count coupling, candidate support, or transfer to Chicago. Multi-seed, size, sparsity, coarsening, and candidate-omission sweeps remain required evidence.

7 Chicago Flagship Application: Evidence Gate

Chicago’s public Transportation Network Provider table reports whether each customer transaction shared a vehicle with a separately booked customer and reports the pooled run size [7, 8]. The provider-side Shared Trip ID groups the empty-to-empty vehicle run but is absent from the public schema. Conditional on run closure and reporting consistency, records with match true and $K = 2$ therefore induce a hidden nonbipartite transaction matching. This is neither a passenger count nor a claim that $K > 2$ runs form co-presence cliques.

Current metadata documents an at-most-two-trip threshold with paired endpoint coarsening [5, 6]. The committed adapter encodes only fail-closed implications. Visible fine geography emits applicable endpoint-marginal HIGH requirements. A blank emits LOW only under content-addressed, metadata-pinned evidence with an independent authority contract; otherwise it contributes no converse threshold claim. All supplied core, buffer, and context rows enter applicable count factors. The adapter verifies count-universe and support digests and rejects node, role, support, namespace, or preloaded-factor drift. Its support-coverage flag concerns only declared per-record latent labels and never candidate-edge coverage. Live extraction and City implementation validation remain false.

The committed $K = 2$ production audit separately requires literal shared-match true, integer pooled size two, an all-row pinned snapshot, and core-buffer boundary handling. Authorization is not a necessary screen. Under the declared nearest-15-minute interpretation it retains pairs whose closed ± 7.5 -minute occupancy envelopes can overlap, including equality and cross-midnight candidates. Missing or malformed times retain otherwise admissible edges while blocking production readiness. Spatial radii, direction rules, and degree caps appear only on a heuristic-sensitivity graph. Small covers are exact; larger HiGHS covers remain numerical and uncertified.

These interfaces establish neither true-partner recall nor hidden-run closure from public rows. The prefix sample is not a closed relation sample and is excluded from estimation. A complete analysis still requires a stabilized all-row snapshot [4], independently

reviewed operator and duration evidence, candidate-support validation, and certified component statuses. No Chicago composition or policy sign is reported.

8 Limitations and Ethical Considerations

The endpoint guarantee is conditional on the declared world set. Candidate omissions, an incorrect release operator, nonexchangeable calibration markets, or unmodeled attribute supports can remove truth before optimization. The split-conformal result is marginal across complete markets, not conditional on every market and not transferable from simulation by assertion. Numerical MILP optima are not exact mathematical certificates; production reporting must retain primal/dual gaps and unresolved states.

The Chicago unit is a customer transaction, not a stable person. Census attributes describe neighborhood context, not rider income or race. Releasing one extremal latent assignment could facilitate reidentification, so the intended artifact reports only aggregate endpoints, support diagnostics, and solver certificates. It must not publish imputed partner links, trip IDs, or latent fine-geography assignments. Institutional review and the City’s data terms must be resolved before any nonpublic data are used.

9 Generative AI Usage

Generative AI assisted drafting, counterexample search, and editing. All claims, experiments, citations, and code remain author-verified; no generated text or output is treated as Chicago evidence.

10 Conclusion

Privacy coarsening can hide a relation and its endpoint attributes jointly, so we optimize feasible worlds rather than one reconstructed graph. The temporal frontier is exact under controlled structural width and score resource; standard incidence decomposition separates only genuine record-factor components. All-ten UCI closes the topology gate but reveals nonmatching truth and an unresolved lower endpoint. Chicago interfaces close contracts, not observation. Submission still requires verified City semantics, stabilized run-closed data, candidate support, and certified production endpoints.

References

- [1] Javier Alonso-Mora, Samitha Samaranyake, Alex Wallar, Emilio Frazzoli, and Daniela Rus. 2017. On-Demand High-Capacity Ride-Sharing via Dynamic Trip-Vehicle Assignment. *Proceedings of the National Academy of Sciences* 114, 3 (2017), 462–467. doi:10.1073/pnas.1611675114
- [2] André Berger, Vincenzo Bonifaci, Fabrizio Grandoni, and Guido Schäfer. 2011. Budgeted Matching and Budgeted Matroid Intersection via the Gasoline Puzzle. *Mathematical Programming* 128, 1–2 (2011), 355–372. doi:10.1007/s10107-009-0307-4
- [3] Maxime Cauchois, Suyash Gupta, Alnur Ali, and John C. Duchi. 2024. Predictive Inference with Weak Supervision. *Journal of Machine Learning Research* 25, 118 (2024), 1–45. <https://jmlr.org/papers/v25/23-0253.html>
- [4] City of Chicago. 2024. Change Notice: Transportation Network Provider Data. Chicago Data Portal, story wmdt-6h9e. Issued 2024-12-12; accessed 2026-08-27. <https://data.cityofchicago.org/stories/s/wmdt-6h9e>
- [5] City of Chicago. 2026. Census Tract Rules for Taxi and TNP Datasets. Chicago Data Portal, story 28mt-8asw. Original clarification dated 2019-07-29; last edited 2026-03-19; accessed 2026-08-27. <https://data.cityofchicago.org/stories/s/28mt-8asw>
- [6] City of Chicago. 2026. How Chicago Protects Privacy in TNP and Taxi Open Data. Chicago Data Portal, story 82d7-i4i2. Last edited 2026-03-27; accessed 2026-08-27. <https://data.cityofchicago.org/stories/s/82d7-i4i2>
- [7] City of Chicago. 2026. TNP Reporting Manual: Trip. Accessed 2026-08-27. <https://chicago.github.io/tnp-reporting-manual/trip/>
- [8] City of Chicago. 2026. Transportation Network Providers—Trips (2025–). Chicago Data Portal, dataset 6dvr-xwnh. Accessed 2026-08-25. <https://data.cityofchicago.org/d/6dvr-xwnh>
- [9] Rina Dechter. 1999. Bucket Elimination: A Unifying Framework for Reasoning. *Artificial Intelligence* 113, 1–2 (1999), 41–85. doi:10.1016/S0004-3702(99)00059-4
- [10] Ilan Doron-Arad, Ariel Kulik, and Hadas Shachnai. 2023. An EPTAS for Budgeted Matching and Budgeted Matroid Intersection via Representative Sets. In *50th International Colloquium on Automata, Languages, and Programming (LIPIcs, Vol. 261)*. 49:1–49:16. doi:10.4230/LIPIcs.ICALP.2023.49
- [11] Su Feng, Boris Glavic, Aaron Huber, and Oliver A. Kennedy. 2021. Efficient Uncertainty Tracking for Complex Queries with Attribute-Level Bounds. In *Proceedings of the 2021 International Conference on Management of Data*. 528–540. doi:10.1145/3448016.3452791
- [12] Matteo Fischetti and Juan José Salazar. 2001. Solving the Cell Suppression Problem on Tabular Data with Linear Constraints. *Management Science* 47, 7 (2001), 1008–1027. doi:10.1287/mnsc.47.7.1008.9805
- [13] Eugene C. Freuder. 1990. Complexity of K -Tree Structured Constraint Satisfaction Problems. In *Proceedings of the Eighth National Conference on Artificial Intelligence*. AAAI Press, 4–9.
- [14] Ming Hua and Jian Pei. 2012. Aggregate Queries on Probabilistic Record Linkages. In *Proceedings of the 15th International Conference on Extending Database Technology*. 360–371. doi:10.1145/2247596.2247639
- [15] Nancy G. Kinnarsley. 1992. The Vertex Separation Number of a Graph Equals Its Path-Width. *Inform. Process. Lett.* 42, 6 (1992), 345–350. doi:10.1016/0020-0190(92)90234-M
- [16] Van Bang Le and Florian Pfender. 2014. Complexity Results for Rainbow Matchings. *Theoretical Computer Science* 524 (2014), 27–33. doi:10.1016/j.tcs.2013.12.013
- [17] Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. 2018. Distribution-Free Predictive Inference for Regression. *J. Amer. Statist. Assoc.* 113, 523 (2018), 1094–1111. doi:10.1080/01621459.2017.1307116
- [18] Oded Maron and Tomás Lozano-Pérez. 1998. A Framework for Multiple-Instance Learning. In *Advances in Neural Information Processing Systems 10*. 570–576.
- [19] Richard (Alex) Mucci and Gregory D. Erhardt. 2024. Understanding Ride-Hailing Sharing and Matching in Chicago Using Travel Time, Cost, and Choice Models. *Transportation Research Record* 2678, 2 (2024), 293–306. See corrigendum: 10.1177/03611981241284953. doi:10.1177/03611981231173636
- [20] Python Record Linkage Toolkit Developers. 2023. Record Linkage Toolkit Datasets, Version 0.16. Accessed 2026-08-27. <https://recordlinkage.readthedocs.io/en/stable/ref-datasets.html>
- [21] Mauricio Sadinle. 2017. Bayesian Estimation of Bipartite Matchings for Record Linkage. *J. Amer. Statist. Assoc.* 112, 518 (2017), 600–612. doi:10.1080/01621459.2016.1148612
- [22] Paolo Santi, Giovanni Resta, Michael Szell, Stanislav Sobolevsky, Steven H. Strogatz, and Carlo Ratti. 2014. Quantifying the Benefits of Vehicle Pooling with Shareability Networks. *Proceedings of the National Academy of Sciences* 111, 37 (2014), 13290–13294. doi:10.1073/pnas.1403657111
- [23] Irene Schmittmann, Gael Hammer, Murat Sariyar, and Aslihan Gerhold-Ay. 2009. Record Linkage Comparison Patterns. UCI Machine Learning Repository. doi:10.24432/C51K6B
- [24] Elaheh Sebt and Ying Chen. 2026. Mining Hidden Ridesharing Patterns: A Data-Driven Gap Analysis of Chicago TNC Trips. *Data Science for Transportation* 8 (2026), 7. doi:10.1007/s42421-026-00149-5
- [25] Jason Soria and Amanda Stathopoulos. 2021. Investigating Socio-Spatial Differences between Solo Ridehailing and Pooled Rides in Diverse Communities. *Journal of Transport Geography* 95 (2021), 103148. doi:10.1016/j.jtrangeo.2021.103148
- [26] Morteza Taiebat, Elham Amini, and Ming Xu. 2022. Sharing Behavior in Ride-Hailing Trips: A Machine Learning Inference Approach. *Transportation Research Part D: Transport and Environment* 103 (2022), 103166. doi:10.1016/j.trd.2021.103166
- [27] Deniz Turkcapar and Sanjay Krishnan. 2023. Quantifying Uncertainty in Aggregate Queries over Integrated Datasets. arXiv:2309.05178 [cs.DB] doi:10.48550/arXiv.2309.05178

A Reproducibility and Claim Status

The artifact includes the complete benchmark generator, split-conformal calibrator, exact temporal-frontier solver, small-instance matching oracle, sparse numerical reference solver, fixed seeds, result tables, and executable adversarial examples. Hidden pair truth is read only in the source-training, calibration, or held-out-test stage declared by the benchmark split. The earlier two-day weak-node-score pilot is retained as failure-analysis provenance but is not a headline result.

Current completed claims comprise Propositions 1, 2, 3, 4, and 5; Lemmas 1 and 2; and Theorems 1, 2, 3, and 4. The artifact has 150 passing tests in the pinned cached environment, eight legacy counterexamples, the 34-case temporal benchmark, the 250-instance MILP audit, 160 same-kernel and 192 independent component-frontier checks, and the controlled calibration results in Table 5. The DNF compiler, Chicago safe adapter, $K = 2$ production audit, bounded capacity profile, all-ten UCI topology audit, and FEBRL4 method-fit audit are complete only under their declared contracts. The UCI lower dyad endpoint, live City operator validation, full-day run closure, true-partner support, and complete-day Chicago estimates remain open.

B Proof of Proposition 3

Let H be the declared record-factor incidence graph. Every selectable edge, degree constraint, allowed label pair, support restriction, count bound, and release requirement lies within one connected component of H . Restricting a structurally feasible global world to the components is therefore well defined. Conversely, the union of arbitrary structurally feasible component worlds satisfies every structural constraint. Restriction and union are inverse, so they give a bijection between global structural worlds and tuples of component worlds.

Omitted-edge use, the single globally shifted score, and the query are additive under this bijection. The global world is feasible exactly when the component omission uses sum to at most Γ and their scores sum to the global floor. For minimization, a local record with no greater omission use, no smaller score, and no larger query can replace a dominated record under every continuation; reverse only the query inequality for maximization. Induction on the convolved components proves that Pareto pruning retains an optimizer for each endpoint. The global score shift is valid because fixed core incidence adds the same constant once per core in every feasible world. Retained local witnesses union to an attained global witness, which is replayed against the unsplit problem. A factor shared by records joins their components in H ; hence the argument does not apply to candidate-graph components alone.

C Full Proof of Theorem 1

Define the decision problem as follows. Records have pickup and drop-off time labels, finite attribute supports, and an observed symbol that is either a visible attribute pair or missing. A world chooses both attributes. The paired operator releases both exactly when both selected endpoint cells have count at least k . Target records form left and right sets and must be perfectly matched; context records cannot be matched. An explicitly represented table $\Gamma_{xy}(p_x)$ determines compatibility. A certificate is polynomially checked by

computing all counts and matching degrees, so the problem is in NP.

Reduce perfect 3D matching with $|X| = |Y| = |Z| = q$. If needed, add one disjoint forced triple so $q \geq 2$ without changing the answer. Create target records r_x and r_y and the complete outer graph $X \times Y$. All r_x share one pickup time and have support Z . Each r_x has a private drop-off time and a singleton drop-off support; its observed pair is missing. Each r_y has private endpoint times, singleton supports, and a missing observed pair.

For each $z \in Z$, create $k - 1$ context anchors with the common pickup time and singleton pickup support $\{z\}$. All anchors share a drop-off time and a singleton label unused by targets, and are visibly released. The anchor drop-off count is $q(k - 1) \geq k$. If $n_z = |\{x : p_x = z\}|$, each z -anchor pickup count is $(k - 1) + n_z$; visibility is therefore equivalent to $n_z \geq 1$. Because $\sum_z n_z = q$, every $n_z = 1$. Missing target outputs are consistent: every x has a unique count-one drop-off cell, and every y has unique count-one endpoint cells.

Finally set $\Gamma_{xy}(z) = 1$ exactly for input triples (x, y, z) . A feasible world matches every x and y once and assigns every z once, so its selected triples form a perfect 3D matching. Conversely, a perfect 3D matching selects the corresponding p_x and (x, y) edges; all anchor counts equal or exceed k , all targets remain suppressed, and every selected edge is compatible. For fixed k the construction has $O(q)$ records, $O(q^2)$ support and outer-edge entries, and at most an $O(q^3)$ explicit compatibility table. The reduction is polynomial.

D Proof Details for Theorem 2

Let X_t be a live key after action t and let $\mathcal{W}_t(X_t)$ be the partial compiled worlds that induce its live labels and matching bits, active capped counts and merged requirements, used omission budget, and capped score. They have already satisfied every forgotten-core degree constraint and every closed factor. For endpoint sense $\circ \in \{\min, \max\}$, the invariant is

$$V_t^\circ(X_t) = \bigcirc_{F \in \mathcal{W}_t(X_t)} H_t(F),$$

with the value attained by the stored partial witness.

The empty layer satisfies the invariant. A node introduction enumerates every compiled label, increments precisely its active factor automata, merges its requirements, and rejects exactly conflicts or factor states with no feasible completion. An edge action enumerates skip and select; select is legal exactly when both matching bits are zero and the live labels form an allowed pair, and it updates the two global resources. A forget action checks a core's bit and removes data whose final incident edge and factor have already been processed. Closing a factor applies its terminal predicate and removes its automaton. Thus every transition is a legal extension and every legal extension is enumerated.

By the schedule's running-intersection property, two partial worlds with the same key have identical future extensions and future additive contributions. Keeping only their attained minimum and maximum is therefore lossless. Induction proves the invariant at every layer. Terminal accepting paths have all core degrees one, all factors closed, omission count at most Γ , and capped score b ; conversely every feasible full world traces such a path. Backtracking yields the two endpoint witnesses.

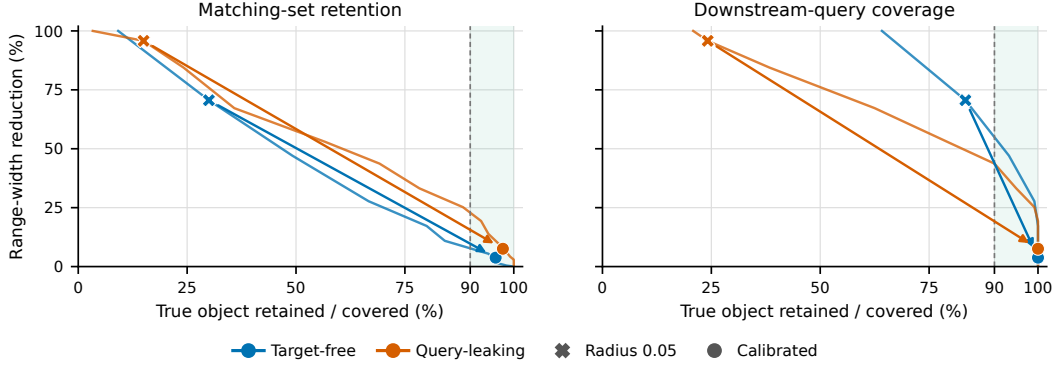


Figure 2: Exact radius frontiers show that the tight stress point has large width reduction but poor retention/coverage; calibration crosses the nominal 90% line while removing most apparent precision. Arrows connect radius 0.05 to the calibrated radius.

For live record i , there are $m_i d'_i$ label/bit assignments; active factor f has at most $3(K_f + 1)$ automaton states, alongside $\Gamma + 1$ omission and $b + 1$ capped-score states. Their product gives Eq. (6); each explicit transition performs polynomial work. The reference implementation copies complete partial witnesses for auditability, whereas a predecessor-pointer implementation attains the usual rolling-state storage bound.

E Full Proof of Theorem 3

Membership in NP follows by checking a proposed perfect matching and its two integer sums. Reduce SUBSET SUM with positive integers $a_1, \dots, a_m$ and target B . For each item create a cycle $u_i - v_i - x_i - y_i - u_i$ with all four vertices core. Its only perfect matchings are

$$M_i^{\text{take}} = \{u_i v_i, x_i y_i\}, \quad M_i^{\text{skip}} = \{v_i x_i, y_i u_i\}.$$

Assign $s_{u_i v_i} = a_i$ and zero score to the other three edges. Since the edge is core-core, taking contributes $2a_i$ to the core-incidence score. Give the same take edge query $2a_i$, with every other query zero. A global perfect matching independently chooses take or skip in each cycle and hence has score and query both equal to $2 \sum_{i \in I} a_i$. Requiring score at least $2B$ and query at most $2B$ is therefore equivalent to $\sum_{i \in I} a_i = B$.

Each cycle has the width-two path decomposition

$$\{u_i, v_i, y_i\}, \quad \{v_i, x_i, y_i\}.$$

Decompositions of disjoint components can be concatenated without increasing width. The associated decision problem is pseudo-polynomially solvable by the score-frontier DP, so it is weakly NP-complete.